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COURSE CODE: ECA0910
COURSE NAME: DIGITAL SIGNAL PROCESSING
PRACTICAL LAB RESULTS

[illegible]

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%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
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```

```
w = zeros(L,1);
y_lms = zeros(N,1);
e_lms = zeros(N,1);
for n=L:N
    xvec = x(n:-1:n-L+1);
    y_lms(n) = w'*xvec;
    e_lms(n) = d(n)-y_lms(n);
    w = w + mu*e_lms(n)*xvec;
```

```
end
```

```
figure;
plot(e_lms,'LineWidth',1);
grid on;
title('LMS Output');
xlabel('Samples');
ylabel('Amplitude');
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
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%% Objective 3 : NLMS
```

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%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
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```
w = zeros(L,1);
y_nlms = zeros(N,1);
e_nlms = zeros(N,1);
for n=L:N
    xvec = x(n:-1:n-L+1);
    y_nlms(n)=w'*xvec;
    e_nlms(n)=d(n)-y_nlms(n);
    w=w+(mu_nlms/(delta+xvec'*xvec))*e_nlms(n)*xvec;
```

```
end
```

```
figure;
plot(e_nlms,'LineWidth',1);
grid on;
title('NLMS Output');
xlabel('Samples');
ylabel('Amplitude');
```

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%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
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%% Objective 4 : MSE Comparison
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```
MSE_LMS=zeros(N,1);
MSE_NLMS=zeros(N,1);
```

```

for k=1:N
    MSE_LMS(k)=mean(e_lms(1:k).^2);
    MSE_NLMS(k)=mean(e_nlms(1:k).^2);
end
figure;
semilogy(MSE_LMS,'r','LineWidth',1.2);
hold on;
semilogy(MSE_NLMS,'b','LineWidth',1.2);
grid on;
legend('LMS','NLMS');
title('MSE Comparison');
xlabel('Iterations');
ylabel('Mean Square Error');

```

OUTPUT:





